

[citation needed] := If this would not be enough elaboration on a coznest, please make a note. Otherwise ignore it.

Shaded = part of a previous graded solution

Problem 1 (17AMO1)

For any prime $p \equiv 1 \pmod{4}$, we claim that $a = \frac{p+1}{2}$; $b = \frac{3p-1}{2}$ works. Note that,

$$\gcd(a, b) = \gcd\left(\frac{p+1}{2}, \frac{3p-1}{2}\right) = \gcd\left(\frac{p+1}{2}, \frac{p-3}{2}\right) = \gcd\left(2, \frac{p-3}{2}\right)$$

Since $p \equiv 1 \pmod{4}$, $\frac{p-3}{2}$ is odd so they are relatively prime. Note that,

$$a + b \mid a^b + b^a \Leftrightarrow a^b + b^a \equiv 0 \pmod{a + b}$$

$$\Leftrightarrow (-b)^b + b^a \equiv 0 \pmod{a + b}$$

$$\Leftrightarrow b^a \equiv b^b \pmod{a + b}$$

Since $\gcd(b, a + b) = 1$,

$$\Leftrightarrow b^{b-a} \equiv 1 \pmod{a + b}$$

We know $a + b = 2p$ and $b - a = p - 1$. Since $\phi(2p) = p - 1$, by Euler's Totient Theorem, since $\gcd(b, a + b) = 1$, this works.

By Dirichlet, there are infinitely many primes $p \equiv 1 \pmod{4}$, so we are done.

Problem 2 (99SLN3)

We let $a_i = p_i^2$ where p_i is the i -th prime that is $1 \pmod{4}$, of which there are infinitely many by Dirichlet. Since they are relatively prime, we independently check that there exists a b_n that is divisible by both $a_n, a_n + 1$.

For,

$$a_n + 1 = p_i^2 + 1 \mid b_n^2 + 1$$

This is clearly possible for $b_n \equiv p_i \pmod{p_i^2 + 1}$. For,

$$p_i^2 \mid b_n^2 + 1$$

By (lemma I don't know), -1 is a quadratic residue mod p_i^2 , so $b_n \equiv a \pmod{p_i^2}$ for some a . By CRT, we have, for some b ,

$$b_n \equiv b \pmod{p_i^2(p_i^2 + 1)}$$

We may then find a sufficiently large $b_n > b_{n-1}$.

Problem 3 (19USMCA1)

We claim that Kelvin may append either a 7 or 8 to the end of the string for every move of the game such that Alex cannot have a move that wins on the next round.

For the first move, Kelvin plays a 7, which is not a square. Alex obviously cannot win on the next move.

For every move after, Kelvin appends either a 7 or 8. Since 7, 8 are not quadratic residues mod 10, the current number is not a square. For the same reason, for Alex to win on the next move, he must append a number to the end of the string.

At this point, there will be 3 numbers on the board after Kelvin and Alex move next with the first being 7. Therefore, it is the square of at least 25, so Alex cannot have a winning response for both 7 and 8 being appended.

Problem 4 (07SLN2)

We choose k such that $v_p(k) > v_p(b)$ for all $p \mid b$. We now have,

$$k \mid b - a_k^n$$

For any prime p ,

$$\Rightarrow v_p(b - a_k^n) \geq v_p(k)$$

$$\Rightarrow v_p(b - a_k^n) > v_p(b)$$

Therefore, $v_p(a_k^n) = nv_p(a_k) = v_p(b)$. This means $n \mid v_p(b)$ for any prime p . Equivalently, $b = A^n$ for some A .

Problem 5 (09APMO4)

The case where $n = 1, 2$ is trivial.

For $n > 2$, we may choose the fractions with value,

$$\frac{n!^2 + 1}{n!}, \frac{n!^2 + 2}{n!}, \dots, \frac{n!^2 + n}{n!}$$

Note that $\gcd(n! + k, n!) = \gcd(k, n!) = k$ for all $1 \leq k \leq n$. Therefore, these respectively simplify to,

$$a_i = \frac{n!^2}{i} + 1; b_i = \frac{n!}{i}$$

We have,

$$a_1 > a_2 > \dots > a_n = n! (n - 1)! + 1 > n! = b_1 > b_2 > \dots > b_n$$

Therefore, all $2n$ numbers are pairwise distinct as desired.

Problem 6 (17SLN2)

We claim Ankan always wins.

The first step is always to plug $a_0 = 0$.

Case 1: $p = 2, 5$

We are already done.

$$N = a_1 \cdot 10 + a_2 \cdot 10^2 + \cdots + a_{p-1} \cdot 10^{p-1} \equiv 0 \pmod{2, 5}$$

Case 2: $2 \mid \text{ord}_p(10)$

Let $k = \text{ord}_p(10)$. Note that,

$$10^k \equiv 1 \pmod{p}$$

$$\Leftrightarrow 10^{\frac{k}{2}} \equiv -1 \text{ or } 1 \pmod{p}$$

The second option is impossible since $\text{ord}_p(10) > \frac{k}{2}$, so $10^{\frac{k}{2}} \equiv -1 \pmod{p}$. Therefore, we split the sum $a_1 \cdot 10 + a_2 \cdot 10^2 + \cdots + a_{p-1} \cdot 10^{p-1}$ into $\frac{p-1}{k}$ groups of k products. In each product, we pair up indices that are $\frac{k}{2}$ apart... (you get the idea)

Case 3: $2 \nmid \text{ord}_p(10)$

We let $k = \text{ord}_p(10)$ and $m = \frac{p-1}{k}$. Since $p-1$ is even but k odd, m is even. We let the k residues when taking powers of 10 modulo p be,

$$r_1, r_2, \dots, r_k$$

On $\{10^1, \dots, 10^{p-1}\}$, there will be m of each, so an even number of each.

Whenever Ryan selects an index i and number a_i , we'd select another index j that leaves the same residue ($10^i \equiv 10^j \pmod{p}$) and set $a_j = 10 - a_i$. In the end, we'd have,

$$\begin{aligned} N &= a_0 + a_1 \cdot 10 + a_2 \cdot 10^2 + \cdots + a_{p-1} \cdot 10^{p-1} \\ &\equiv 0 + \frac{10}{2} \cdot m(r_1 + r_2 + \cdots + r_k) \pmod{p} \end{aligned}$$

Note that,

$$r_1 + r_2 + \cdots + r_k = 10 + 10^2 + \cdots + 10^k = 10 \cdot \frac{10^k - 1}{10 - 1} \equiv 0 \pmod{p}$$

Therefore, $N \equiv 0 \pmod{p}$.

+4 clubs; 2 hours

Problem 7 (23USEMO1)

Couldn't do this at all in the contest and solved this today in 20 min lol.

Define a_b to be a base b number.

Lemma: For a pair of integers a, b, n, m , where $k = \gcd(n, m)$, we may find infinitely integers x that satisfy $x \equiv a \pmod{n}$ and $x \equiv b \pmod{m}$ if $a \equiv b \pmod{k}$.

Proof: Same as proof of CRT.

Call a number k -beautiful for some $k \geq 4$ if, for every $4 \leq b \leq k$, the base- b representation of the number has digits 2, 0, 2, 3 in order.

We prove that, for all $k \geq 4$, we may find distinct numbers a, n such that all numbers of the form $a \pmod{n}$ are k -beautiful. We do this with induction.

Base case: $k = 4$

All numbers that are $2023_4 \pmod{10000_4}$ satisfy this.

Inductive step: $k = x \rightarrow k = x + 1$

By the inductive hypothesis, for some a, n , we have all numbers that are $a \pmod{n}$ are x -beautiful.

Let y be a positive integer such that $(x + 1)^y > n$. Clearly, all numbers which are,

$$(2023_{x+1}) \cdot (x + 1)^y + m \pmod{(x + 1)^{y+4}}$$

For $0 \leq m < (x + 1)^y$ are $x + 1$ -beautiful since the $y + 1, y + 2, y + 3, y + 4$ digits from the right are 2023 in that order.

Note that $\gcd(n, (x + 1)^y) \leq n < (x + 1)^y$, therefore, by Pigeonhole, there exists a m on this range such that,

$$(2023_{x+1}) \cdot (x + 1)^y + m \equiv a \pmod{\gcd(n, (x + 1)^y)}$$

Using this selection of m , we apply the lemma to get for some a', n' , all numbers that are $a' \pmod{n'}$ are still x -beautiful and now additionally $(x + 1)$ -beautiful.

Conclusion: By mathematical induction, we are done.

Hence, there exists an infinite number of 10000-beautiful numbers as desired.

Problem 8 (19CHN2)

Denote $X \leftrightarrow Y$ to indicate that triples X, Y can be linked. This is obviously commutative.

In any Pythagorean triple, there is always a multiple of 4, so we can link,

$$P \leftrightarrow (3p, 4p, 5p), Q \leftrightarrow (3q, 4q, 5q)$$

We prove that we can link any $(3n, 4n, 5n)$ to $(3, 4, 5)$, which is sufficient since then,

$$(3p, 4p, 5p) \leftrightarrow (3, 4, 5) \leftrightarrow (3q, 4q, 5q)$$

We now denote $T(x) = (3x, 4x, 5x)$. Once again, we prove that $T(1) \leftrightarrow T(n)$ for any n .

For any odd prime p , we may link,

$$\begin{aligned} (3p, 4p, 5p) &\leftrightarrow (4p, 2(p^2 - 1), 2(p^2 + 1)) \leftrightarrow \left(3 \cdot \frac{p^2 - 1}{2}, 2(p^2 - 1), 5 \cdot \frac{p^2 - 1}{2}\right) \\ &\Rightarrow T(p) \leftrightarrow T\left(\frac{p^2 - 1}{2}\right) \end{aligned}$$

Therefore, if we scale everything by any integer m ,

$$T(mp) \leftrightarrow T\left(m \cdot \frac{p^2 - 1}{2}\right)$$

Let q be the largest prime factor of n , while $q > 2$, we may do,

$$T(n) \leftrightarrow T\left(\frac{n}{q} \cdot \frac{q^2 - 1}{2}\right)$$

We now let $n := \frac{n}{q} \cdot \frac{q^2 - 1}{2}$.

Since $q^2 - 1 = (q - 1)(q + 1)$, and each factor is an even number which cannot have a prime factor greater than or equal to q . If we do this enough, the multiplicity of q in the prime factorization of n will decrease to zero with no larger primes appearing.

Eventually, the largest prime divisor of n will be 2, meaning $n = 2^k$ for some k .

Claim: $T(2n) \leftrightarrow T(n)$ for any n .

Proof: $(3n, 4n, 5n) \leftrightarrow (5n, 12n, 13n) \leftrightarrow (9n, 12n, 15n) \leftrightarrow (8n, 15n, 17n) \leftrightarrow (6n, 8n, 10n)$

Therefore, inducting, $T(1) \leftrightarrow T(2^k)$ and we are done.

Problem 9 (13SLN3)

We can write $P(n) = n^4 + n^2 + 1$, and $F(n) = n^2 + n + 1$. Note that,

$$P(n) = F(n - 1) F(n);$$

$$P(n + 1) = F(n) F(n + 1)$$

For the sake of contradiction, let there not be infinite n such that $P(n)$ and $P(n + 1)$ do not share a maximum prime divisor.

Therefore, after a certain finite threshold k , all integers $n \geq k$ satisfy $P(n)$ and $P(n + 1)$ not sharing a maximum prime divisor.

We let $G(n)$ be the maximum prime divisor of $F(n)$. We clearly have the maximum prime divisor of $P(n)$ is $\max(G(n - 1), G(n))$. Therefore, for all $n \geq k$,

$$\max(G(n - 1), G(n)) \neq \max(G(n), G(n + 1))$$

We cannot have a scenario where both $G(n) \geq G(n - 1)$ and $G(n) \geq G(n + 1)$.

Claim: $G(n) \neq G(n + 1)$.

Note that,

$$\gcd(F(n), F(n + 1)) = \gcd(n^2 + n + 1, n^2 + 3n + 3) = \gcd(n^2 + n + 1, 2n + 2)$$

Since $n^2 + n + 1 = n(n + 1) + 1$ is odd,

$$= \gcd(n^2 + n + 1, n + 1) = \gcd(1, n + 1) = 1$$

The two are relatively prime and hence cannot share the same maximum prime factor.

Claim: If, for any $m \geq k$, $G(m) \geq G(m - 1)$, then G is strictly increasing after m .

This is just induction. [citation needed?]

Therefore, we see that $G(n)$ will initially strictly decrease after $n = k$ for some duration, then become strictly increasing. Obviously, since there are a finite number of positive primes, $G(n)$ cannot strictly decrease forever, so, inevitably, there will be a threshold m such that $G(n + 1) > G(n)$ for all $n \geq m$.

[I was planning on using a density argument using the fact that $p_n \sim n \log n$ hoping to prove that eventually $f(n) < p_n$. Clearly this does not work out since $f(n) \sim n^2 > n \log n$, but would it be rigorous enough to use on a proof if it had worked out? Also, can we even state without proof that $p_n \sim n \log n$?]

However, note that,

$$\begin{aligned} F(x^2) &= P(x) = F(x)F(x - 1) \\ \Rightarrow G(x^2) &= \max(G(x), G(x - 1)) \end{aligned}$$

Therefore, G cannot be past a certain point. (!)

+10 clubs; 6 hours

[Problem 10 \(87IMO5\)](#)

We start with the lattice points, (x, x^2) for integers x in range $0 \leq x \leq 1986$. Namely, the points,

$$(0, 0); (1, 1); (2, 4); (3, 9); \dots; (1986, 1986^2)$$

Since the graph of $f(x) = x^2$ is convex, its derivative is monotonic increasing. Therefore, no three distinct points on the graph may be collinear (*by mean value theorem*).

Therefore, by shoelace, the current area of every triangle is a positive rational.

We may use the distance formula to write the distance between every pair of points as a not-necessarily-simplified radical $\sqrt{n}$. Note that the distance is maximized between $(0, 0)$ and $(1986, 1986^2)$, so,

$$n \leq 1986^2 + 1986^4$$

To get our construction, we perform a dilation about the origin with scale $\sqrt{p}$ for some prime $p > 1986^2 + 1986^4$.

The area of every triangle is now $(\sqrt{p})^2 = p$ times what it used to be, so still rational.

The distance between every pair of points may now be expressed as $\sqrt{np}$. Since $p > n > 0$, we cannot have $p \mid n$. Therefore, $\sqrt{np}$ must be irrational. [citation needed?]

Problem 11 (Z631B298)

We let $d(x)$ be the sum of the exponents in the prime factorization of x . We claim that the answer is that it is possible if and only if $k \leq p(n + 1)$.

We first prove that it is possible if $k \leq p(n + 1)$.

Claim: If (n, k) works, then $(n, k - 1)$ works for $k > 1$.

We can keep $A_1, \dots, A_{k-2}$ as is, then for the last set, we have it be $A_{k-1} \cup A_k \cup S$ where S is the set of the all possible sums taking one element of A_{k-1} and one element of A_k .

For example, for $(23, 4)$, we could have done $\{1\}, \{2\}, \{4\}, \{8, 16\}$

To create $(23, 3)$, we can do $\{1\}, \{2\}, \{4, 8, 16, 12, 20\}$.

Therefore, consider the case where $k = d(n + 1)$. We write,

$$n + 1 = p_1 p_2 \dots p_k$$

For primes $p_1 \dots p_k$ (repetition allowed). We may start with,

$$A_1 = \{1, 2, \dots, p_1 - 1\}$$

This uniquely generates all the numbers from 1 to $p_1 - 1$ with $\max(A_1) = p_1 - 1$. We next have,

$$A_2 = \{p_1, 2p_1, \dots, (p_2 - 1)p_1\}$$

This would uniquely generate all the numbers from 1 to $p_1 p_2 - 1$ [citation needed?] with $\max(A_1) + \max(A_2) = p_1 p_2 - 1$. We can continue with,

$$A_m = \{(1)p_1 p_2 \dots p_{m-1}, (2)p_1 p_2 \dots p_{m-1}, \dots, (p_m - 1)p_1 p_2 \dots p_{m-1}\}$$

Inductively [citation needed?], this would generate all the numbers from 1 to $p_1 \dots p_m - 1$ with $\max(A_1) + \max(A_2) + \dots + \max(A_m) = p_1 p_2 \dots p_m - 1$.

Continuing until A_k , we'd have k sets which uniquely generates the numbers from 1 to $p_1 \dots p_k - 1 = n$ with maximum n as well.

We may now prove that no greater k works.

Claim: In a working pair (n, k) with working sets $A_1 \dots A_k$,

$$n + 1 = (|A_1| + 1)(|A_2| + 1) \dots (|A_k| + 1)$$

For each set A_m , we can either choose one of the $|A_m|$ numbers in the set to add to or total or not choose anything. In total, we'd have $(|A_1| + 1)(|A_2| + 1) \dots (|A_k| + 1)$ ways. However, we take one away for the unpermitted special case where we take nothing. The other $(|A_1| + 1)(|A_2| + 1) \dots (|A_k| + 1) - 1$ options should uniquely generate the numbers from 1 to n , so we are done.

For the sake of contradiction, let $k > p(n + 1)$. We'd have to find k numbers each at least two which multiply to $n + 1$. This is clearly impossible since at best we can break $n + 1$ into its prime factors.

Problem 12 (11BAMO5)

It is possible.

We have one pair off by one and one pair off by two. For,

$$\begin{aligned} \frac{\binom{n}{k}}{\binom{n}{k+1}} &= \frac{1}{2} \\ \Leftrightarrow \frac{\frac{n!}{k!(n-k)!}}{\frac{n!}{(k+1)!(n-k-1)!}} &= \frac{k+1}{n-k} = \frac{1}{2} \end{aligned}$$

We may have $k = \frac{n-2}{3}$. Next, for the off by twos,

$$\begin{aligned} \frac{\binom{n}{k}}{\binom{n}{k+2}} &= \frac{1}{2} \\ \Leftrightarrow \frac{\frac{n!}{k!(n-k)!}}{\frac{n!}{(k+2)!(n-k-2)!}} &= \frac{(k+1)(k+2)}{(n-k)(n-k-1)} = \frac{1}{2} \\ \Leftrightarrow 2(k+1)(k+2) &= (n-k-1)(n-k) \\ \Leftrightarrow 2\left(k + \frac{3}{2}\right)^2 - \frac{1}{2} &= \left(n - k - \frac{1}{2}\right)^2 - \frac{1}{4} \end{aligned}$$

$$\Leftrightarrow 2(2k+3)^2 - 1 = (2n - 2k - 1)^2$$

$$\Leftrightarrow -1 = (2n - 2k - 1)^2 - 2(2k+3)^2$$

We first solve the negative Pell's equation $-1 = a^2 - 2b^2$, which has solution $(1, 1)$ corresponding to $1 + \sqrt{2}$. Looking at,

$$(1 + \sqrt{2})^3 = 7 + 5\sqrt{2} \rightarrow (a, b) = (7, 5) \rightarrow (n, k) = (5, 1)$$

This doesn't work because it's the same numbers as off-by-one.

$$(1 + \sqrt{2})^5 = 41 + 29\sqrt{2} \rightarrow (a, b) = (41, 29) \rightarrow (n, k) = (34, 13)$$

n is not 2 mod 3, so there is no off-by-one.

$$(1 + \sqrt{2})^7 = 239 + 169\sqrt{2} \rightarrow (a, b) = (239, 169) \rightarrow (n, k) = (203, 83)$$

This works. Indeed,

$$\binom{203}{85} = 2 \binom{203}{83};$$

$$\binom{203}{68} = 2 \binom{203}{67}$$

$$(1 + \sqrt{2})^3 = 7 + 5\sqrt{2}$$

+18 clubs; 8 hours

Problem 13 (12TSTST5)

Note that for $b_{n+1}, \dots, b_m \in \{0, 1\}$, we may write, for all $m > n$,

$$x_m = 2^{m-n}x_n + \frac{2^{m-n-1}}{n+1}b_{n+1} + \frac{2^{m-n-2}}{n+2}b_{n+2} + \dots + \frac{2^0}{m}b_m$$

We can easily prove this by induction.

Claim: Starting at any n , x_n , we may clear an odd prime factor p in the denominator of x_n replacing it with a power of two.

We let $q = p^e \parallel x_n$. [valid notation?] We force all $b_x = 0$ if $x \neq 2^k q$. Therefore, we are left with, for some r, s such that $n < 2^r q < 2^s q < m$,

$$\begin{aligned} x_m &= 2^{m-n}x_n + \frac{2^{m-2^r q}}{2^r q}b_{2^r q} + \frac{2^{m-2^{r+1} q}}{2^{r+1} q}b_{2^{r+1} q} + \dots + \frac{2^{m-2^s q}}{2^s q}b_{2^s q} \\ &= \frac{2^{s+m-n}x_n}{2^s q} + \frac{2^{(s-r)+m-2^r q}}{2^s q}b_{2^r q} + \frac{2^{(s-r-1)+m-2^r q}}{2^s q}b_{2^{r+1} q} + \dots + \frac{2^{m-2^s q}}{2^s q}b_{2^s q} \end{aligned}$$

For convenience, we write set the coefficients to $c_1, \dots, c_{s-r+1}$.

$$= \frac{2^{s+m-n}x_n + c_1b_{2^r q} + c_2b_{2^{r+1}q} + \cdots + c_{s-r+1}b_{2^s q}}{2^s q}$$

Note that no matter what $c_1, \dots, c_{s-r+1}$ are, the set $\{0, c_1\} + \{0, c_2\} + \{0, c_3\} + \cdots + \{0, c_m\}$ spans everything modulo p since c_i is a power of two and cannot be a multiple of p . Taking m large enough, we can get the numerator to be a multiple p . Simplifying the fraction, this eliminates a factor of p in the denominator.

With the claim proven, we just replace all the odd prime factors and be left with $x_n = \frac{m}{2^k}$. We then just double it until $x_{n+k} = m$ to be done.

Problem 14 (14SLN2)

If n is odd, we just take $k = n^m$ for any positive integer m .

$$\left\lfloor \frac{n^{n^m}}{n^m} \right\rfloor = n^{n^m - m}$$

This is clearly odd.

We now look at the case if n is even.

Lemma: For any pair of integers x, y such that $2x < y$, there exists some prime p such that $xp < y < 2xp$

For the sake of contradiction, let this be impossible. This means that there exists no prime p in the range,

$$\frac{y}{2x} < p < \frac{y}{x}$$

Since $\frac{y}{2x} > 1$, this contradicts with a known theorem which states for any real number $n > 1$, there must be a prime p such that $n < p < 2n$.

I forgot the name Bertrand's Postulate when doing the writeup. But is this famous enough that this description would have been accepted?

For any integer $m \geq 3$, we take, a prime p such that,

$$2^m p < n^{2^m} < 2^{m+1} p$$

Clearly, $2^m < n^{2^m}$ for $n > 1$. We then may take $k = 2^m p$. Note that $\phi(k) = 2^{m-1}(p-1)$, and,

$$k \equiv 2^m p - 2\phi(k) = 2^m \pmod{\phi(k)}$$

This way,

$$n^k = n^{2^m p} \equiv n^{2^m} \equiv n^{2^m} - 2^m p \pmod{2^m p}$$

Note that $0 < n^{2^m} - 2^m p < 2^m p$, so,

$$\begin{aligned}
\left\lfloor \frac{n^k}{k} \right\rfloor &= \frac{n^{2^m p} - (n^{2^m} - 2^m p)}{2^m p} \\
\Rightarrow v_2 \left(\left\lfloor \frac{n^k}{k} \right\rfloor \right) &= v_2 \left(\frac{n^{2^m p} - (n^{2^m} - 2^m p)}{2^m p} \right) \\
&= v_2 (n^{2^m p} - (n^{2^m} - 2^m p)) - m
\end{aligned}$$

Since $v_2(n^{2^m p}), v_2(n^{2^m}) > v_2(2^m p)$, we have $v_2(n^{2^m p} - (n^{2^m} - 2^m p)) = v_2(2^m p) = m$. Therefore, $v_2 \left(\left\lfloor \frac{n^k}{k} \right\rfloor \right) = 0$ and we are done with $\left\lfloor \frac{n^k}{k} \right\rfloor$ odd.

+24 clubs; 11 hours

Problem 15 (07USATST4)

We claim the answer is no.

Claim: For any fixed integers (a, b) , the sequence $b, b^b, b^{b^b}, \dots$ will become constant modulo a .

Note that for a sufficiently large x , $b^{x+\phi(b)} = b^x \pmod{a}$. Therefore, we may prove that $b, b^b, b^{b^b}, \dots$ will become constant modulo $\phi(a)$. We use induction to get eventually we have $\phi(\phi(\dots \phi(a))) = 1$, by which time we'll be done.

Problem 16 (18CHNTST24)

Since $k - 1$ is not square free, we let some $p^2 \mid k - 1$.

We work partially in base k and choose, for $d = \frac{k-1}{p}$ and some integer N ,

$$\alpha = N + \frac{1}{p} = N + 0.\overline{ddd \dots}_k$$

Therefore,

$$\begin{aligned}
\lfloor \alpha k^n \rfloor &= k^n N + \overline{ddd \dots}_k = k^n N + d \cdot \frac{k^n - 1}{k - 1} = k^n N + \frac{k^n - 1}{p} \\
&= \left(N + \frac{1}{p} \right) k^n - \frac{1}{p}
\end{aligned}$$

We pick N such that $p \nmid N$. Since $\lfloor \alpha k^n \rfloor = k^n N + d \cdot \frac{k^n - 1}{k - 1}$, and $p \mid d$, we'd have $p \nmid \lfloor \alpha k^n \rfloor$.

For any other prime $q \mid M$, we pick N such that $q \mid N + p^{-1}$. This would mean q divides $\left(N + \frac{1}{p} \right)$ but obviously not $\frac{1}{p}$, so we'd have $q \nmid \lfloor \alpha k^n \rfloor$.

By the Chinese remainder theorem, this is possible and we are done.

Problem 17 (18EGMO2)

For any odd n , $3 \mid 2^n + 1$. Therefore, we may choose $x = 3; y = \frac{2^n+1}{3}$ for any $n > 3$, of which there are infinitely many,

Note that, applying $1 + \frac{1}{2^n}$ then applying the 2 multiplication n times.

$$f(2^n + 1) \leq f(2^n) + 1 \leq n + 1$$

Additionally, we also obviously have,

$$f(x) = f(3) = 2$$

$$\text{Claim: } f\left(\frac{2^n+1}{3}\right) \geq n$$

FTSOC, let this be attainable in $n - 1$ moves or less. Note that we must choose 2 at least $n - 2$ times for if not, the maximum we can attain is obviously,

$$\begin{aligned} 2^{n-3} \cdot \left(\frac{3}{2}\right)^2 &< \frac{2^n + 1}{3} \\ \Leftrightarrow 2^{n-3} \cdot \frac{27}{4} &< 2^n + 1 \quad (!) \end{aligned}$$

Therefore, for some x ,

$$\frac{2^n + 1}{3} = 2^{n-2} \cdot \left(\frac{x+1}{x}\right)$$

However, since $2^n + 1$ is odd, the RHS must also be an odd integer. The only way this is possible is if $v_2(x) = n - 2$. Since $x \leq 2^{n-2}$, we have $x = 2^{n-2}$. Therefore,

$$\begin{aligned} \frac{2^n + 1}{3} &= 2^{n-2} + 1 \\ \Leftrightarrow 2^n &= 3 \cdot 2^{n-2} + 2 \\ \Leftrightarrow 2^{n-2} &= 2 \end{aligned}$$

For $n > 3$, this is clearly impossible.

With the claim proven, we hence have, for this selection of x, y , of which there are infinitely many,

$$f(xy) \leq n + 1 < f(x) + f(y)$$

Problem 18 (15BRA3)

(I got the 27, 169 by myself :D but couldn't figure out square-free)

We claim all n of the form,

$$n = 27(169 \cdot 78t - 25)$$

Where $169 \cdot 78t - 25$ and $27 \cdot 78t - 4$ are square-free numbers work.

Note that, for any square free integer y , we obviously have $f(y) = 1$. Additionally, f is obviously multiplicative [citation needed?]. Since $169 \cdot 78t - 25$ is square free and relatively prime to 3 and hence 27,

$$f(n) = f(27) \cdot f(169 \cdot 78t - 25) = f(3^3) = 27$$

On the other hand,

$$f(n - 1) = f(27 \cdot 169 \cdot 78t - 27 \cdot 25 - 1) = f(169(27 \cdot 78t - 4))$$

Clearly, $27 \cdot 78t - 4$ is relatively prime to 13 and hence 169. Therefore,

$$= f(169) \cdot f(27 \cdot 78t - 4) = 26$$

Therefore, $f(n) = f(n - 1) + 1$ as desired.

We now check that there are infinitely many t which satisfy this condition that $169 \cdot 78t - 25$ and $27 \cdot 78t - 4$ are both square-free.

Claim: For a fixed number M , there are at most $\sum_{p=5; p \text{ is prime}}^{\sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}} \left(2 \left\lfloor \frac{M}{p^2} \right\rfloor\right)$ number of integers t on $1 \leq t \leq M$ such that at least one of $27 \cdot 78t - 4$ and $169 \cdot 78t - 25$ is not square-free.

Consider what the squares primes may divide $27 \cdot 78t - 4$ or $169 \cdot 78t - 25$. Obviously, if,

$$p^2 > \max(27 \cdot 78t - 4, 169 \cdot 78t - 25)$$

This is impossible. For all other primes p , we'd want $27 \cdot 78t \equiv 4 \pmod{p^2}$ or $169 \cdot 78t \equiv 25 \pmod{p^2}$. This is impossible if $p = 13$ or $p = 3$. For all other p , we may respectively take modular inverse and simplify that to $t \equiv x \pmod{p^2}$ and $t \equiv y \pmod{p^2}$ respectively for some not-necessarily-distinct x, y .

Since we are only taking the upper bound, we are not concerned about overcounting, which would reduce the actual maximum. Therefore, we have at most $2 \left\lfloor \frac{M}{p^2} \right\rfloor$ for this prime. Using the same argument, we may just sum these values over all the primes from 5 to $\sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}$. [citation needed?]

However,

$$\begin{aligned} & \sum_{p=5; p \text{ is prime}}^{\sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}} \left(2 \left\lfloor \frac{M}{p^2} \right\rfloor\right) \\ & < \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + \sum_{p=5; p \text{ is prime}}^{\infty} \left(\frac{2M}{p^2}\right) \end{aligned}$$

$$\begin{aligned}
&< \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + 2M \sum_{p=5}^{\infty} \left(\frac{1}{p^2} \right) \\
&= \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + 2M \cdot \left(\frac{\pi^2}{6} - 1 - \frac{1}{4} - \frac{1}{9} - \frac{1}{16} \right) \\
&< \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + \frac{M}{2}
\end{aligned}$$

Therefore, using complementary counting, the amount of t such that they are both square-free.

$$M - \sum_{p=5; p \text{ is prime}}^{\sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}} \left(2 \left\lfloor \frac{M}{p^2} \right\rfloor \right) > \frac{M}{2} - \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}$$

Taking M large, this will approach infinity.

Problem 19 (12RMM4)

We define the following sequence $n_0, n_1, \dots$ as follows.

We start with $n_0 = 3$. For $k \geq 1$, we have $n_k = pn_{k-1}$ where p is the primitive prime divisor of,

$$2^{2^{n_{k-1}+1}} + 1$$

This is possible by Zsigmondy noting that $2^{n_{k-1}+1} > 3$.

We claim every term in this sequence works. We prove this using induction.

Base Case: $n = 3$, we may verify this.

Inductive Step:

Note that,

$$2^{2^{n_k+1}} + 1 = 2^{2^{pn_{k-1}+1}} + 1 = \left(2^{2^{n_{k-1}+1}} \right)^{2^{(p-1)n_{k-1}}} + 1 \equiv (-1)^{2^{(p-1)n_{k-1}}} + 1 \equiv 0 \pmod{2^{2^{n_{k-1}+1}} + 1}$$

Therefore, using the inductive hypothesis,

$$\begin{aligned}
pn_{k-1} &\mid 2^{2^{n_{k-1}+1}} + 1 \\
&\Rightarrow n_k \mid 2^{2^{n_k+1}} + 1
\end{aligned}$$

On the other hand, p does not divide $2^{n_k} + 1$ as,

$$2^{n_k} + 1 \equiv 2^{pn_{k-1}} + 1 \equiv 2^{n_{k-1}} + 1 \pmod{p}$$

Since $n_{k-1} < 2^{n_{k-1}} + 1$, and p is a primitive prime divisor of that, we are done.

+30 clubs; 13 hours

Problem 20 (21RMM2)

We claim the answer is 2.

The construction is by sending $\{17, 19\}$ then $\{19, 20\}$. We show that this can determine $N \bmod 17, 19, 20$.

The union of these two sets is $a_{19} \equiv N \bmod 19$. We see the other distinct element in each set is $a_k \equiv N \bmod k$. Once we have $N \bmod 17, 19, 20$, we may use CRT to find $N \bmod 6460$, which would tell us N as we have $N \leq 5000$.

We now prove that 1 does not work. Namely, we show that $N = 1$ and $N = 2521$ can simultaneously be solutions.

Consider the numbers 11, 13, 16, 17, 19 *big* and other fifteen *small*. For any set $S = \{s_1, \dots, s_n\}$, we return $\{t_1, \dots, t_n\}$. For all i ,

$$t_i \equiv 1 \bmod s_i$$

Additionally, if s_i is big, we force $t_{i+1} \equiv 2521 \bmod s_i$ (indices modulo n). This is possible by CRT since s_i is prime and relatively prime to s_{i+1} . In this case, both $N = 1$ and $N = 2521$ would be solutions.

For $N = 1$, then $a_{s_i} = t_i$. For $N = 2521$, then $a_{s_i} = t_{i+1}$.

Problem 21 (12AMO3)

We claim that all $n > 2$ work.

$n = 2$ fails since we'd need,

$$a_1 + 2a_2 = 0;$$

$$a_2 + 2a_4 = 0;$$

...

Therefore, $a_1 = (-2)^k a_{2^k}$ meaning a_1 cannot be a finite nonzero integer.

For $n > 2$, we show that is possible to come up with a sequence $a_1, a_2, \dots$ such that,

$$a_1 + 2a_2 + 3a_3 + \dots + na_n = 0$$

$$a_{ij} = a_i a_j \quad \forall i, j \geq 1$$

This way, we may factor the a_k out of,

$$a_k + 2a_{2k} + 3a_{3k} + \dots + na_{nk} = 0$$

To finish.

The construction is as follows, for some function f mapping primes to nonzero integers, we may let,

$$a_{p_1^{k_1} p_2^{k_2} \dots p_n^{k_n}} = f(p_1)^{k_1} \dots f(p_n)^{k_n}$$

This would ensure that all terms are nonzero and a is completely multiplicative.

Case 1: $n = 4$

We have $f(3) = -7, f(2) = 2$, and $f(p) = 1$ for all other primes p . Therefore,

$$\begin{aligned} a_1 + 2a_2 + 3a_3 + 4a_4 &= 0 \\ \Leftrightarrow 1 + 2(2) + 3(-7) + 4(4) &= 0 \end{aligned}$$

Case 2: All other n

Claim: For all positive integers n except 1, 2, 4, there are two primes in the range $(\sqrt{n}, n]$.

$$n = 3 \rightarrow 2, 3; n = 5 \rightarrow 3, 5; n = 6 \rightarrow 3, 5$$

For $16 > n \geq 7$, we may take 5, 7.

For $n \geq 16$, there is a prime between $\frac{n}{2}$ and n as well as another prime between $\frac{n}{4}$ and $\frac{n}{2}$ by Bertrand. Since $\sqrt{n} \geq \frac{n}{4}$ we are done.

We let the greatest prime less than or equal to n be p . Note that $\frac{n}{2} < p \leq n$ by Bertrand.

Let a prime other than p that is greater than $\sqrt{n}$ be q .

For all primes p' apart from p, q , we let $f(p') = 1$. Therefore, we find an appropriate $f(p), f(q)$ such that, for some constant M which would be the sum of positive integers less than or equal to n that are relatively prime to both p and q ,

$$\left(qa_q + 2qa_{2q} + \dots + \left\lfloor \frac{n}{q} \right\rfloor qa_{\left\lfloor \frac{n}{q} \right\rfloor q} \right) + pa_p + M = 0$$

Note that there is no $2pa_{2p}$ since $\frac{n}{2} < p$. Since $a_1 = a_2 = \dots = a_{\left\lfloor \frac{n}{q} \right\rfloor} = 1$,

$$\Leftrightarrow \left(qa_q + 2qa_q + \dots + \left\lfloor \frac{n}{q} \right\rfloor qa_q \right) + pa_p + M = 0$$

Since $q^2 > n \Rightarrow pq > n$,

$$\begin{aligned} \Leftrightarrow \left(q + 2q + \dots + \left\lfloor \frac{n}{q} \right\rfloor q \right) (a_q) + pa_p + M &= 0 \\ \Leftrightarrow \frac{\left\lfloor \frac{n}{q} \right\rfloor \left(\left\lfloor \frac{n}{q} \right\rfloor + 1 \right)}{2} q(a_q) + pa_p + M &= 0 \\ \Leftrightarrow \frac{\left\lfloor \frac{n}{q} \right\rfloor \left(\left\lfloor \frac{n}{q} \right\rfloor + 1 \right)}{2} qf(q) + M &= -pa_p \end{aligned}$$

Since $p \nmid \left\lfloor \frac{n}{q} \right\rfloor, \left\lfloor \frac{n}{q} \right\rfloor + 1, q$ as $0 < \left\lfloor \frac{n}{q} \right\rfloor, q, \left\lfloor \frac{n}{q} \right\rfloor + 1 < p$, for some suitable $f(q)$, we can make the LHS a nonzero multiple of p . This would then give us a possible $a_p = f(p)$.

Problem 23 (13SLN4)

We claim that this is impossible.

FTSOC, let there be such a sequence such that for some k , the numbers $\overline{a_k a_{k-1} \dots a_0}, \overline{a_{k+1} a_k \dots a_0}, \overline{a_{k+2} a_{k+1} \dots a_0}, \dots$ are all perfect squares. We let,

$$x_k^2 = \overline{a_k a_{k-1} \dots a_0};$$

$$x_{k+1}^2 = \overline{a_{k+1} a_k \dots a_0};$$

...

We hence have,

$$x_{k+1}^2 - x_k^2 = (x_{k+1} - x_k)(x_{k+1} + x_k) = a_{k+1} \cdot 10^{k+1};$$

$$x_{k+2}^2 - x_{k+1}^2 = (x_{k+2} - x_{k+1})(x_{k+2} + x_{k+1}) = a_{k+2} \cdot 10^{k+2};$$

...

Case 1: $v_5(x_k^2) = 2e < k + 1$

In that case, for $v_5(x_{k+1}^2 - x_k^2) \geq k + 1 > v_5(x_k^2)$, we must have $v_5(x_{k+1}^2) = 2e$ too. Inducting, we have $2e = v_5(x_k^2) = v_5(x_{k+1}^2) = v_5(x_{k+2}^2) = \dots$

Therefore, since exactly one may be more than e ,

$$\min(v_5(x_{k+1} - x_k), v_5(x_{k+1} + x_k)) = e;$$

We then have,

$$\max(v_5(x_{k+1} - x_k), v_5(x_{k+1} + x_k)) = k + 1 - e;$$

Therefore, (I was following solutions and I don't understand the logic of this part, I've asked in OTIS Discord, but nobody has answered yet. Wouldn't it be $\max(x_{k+1} - x_k, x_{k+1} + x_k) \geq 5^{k+1-e}$?)

$$\min(x_{k+1} - x_k, x_{k+1} + x_k) \geq 5^{k+1-e}$$

For sufficiently large k , we have this is greater than $\sqrt{9 \cdot 10^{k+1}}$. Since $(x_{k+1} - x_k)(x_{k+1} + x_k) \leq 9 \cdot 10^{k+1}$, this is a contradiction.

Case 2: $v_5(x_k^2) \geq k + 1$

We similarly use induction to find $v_5(x_k^2) \geq k + 1$ for all sufficiently large k . Additionally, inducting backwards gives $v_5(x_k^2) \geq k + 1$ for all k . Therefore, $v_5(x_0^2) \geq 1$, which is possible only if $x_0 = 25$. Thus, all x_k are odd.

$$(x_{k+1} - x_k)(x_{k+1} + x_k) = a_{k+1} \cdot 10^{k+1}$$

Since x_k is odd, $\min(v_2(x_{k+1} - x_k), v_2(x_{k+1} + x_k)) = 1$ so one is divisible by 2^k . Therefore,

$$\min(x_{k+1} - x_k, x_{k+1} + x_k) \geq 2^k \cdot 5^{\frac{k+1}{2}}$$

Once again, this is larger than $\sqrt{9 \cdot 10^k}$ for sufficiently large k .

Both cases give contradictions, so we are done.

Problem 24 (20AUS8)

We may first let $b_i = x^{\frac{2 \cdot 3 \cdot \dots \cdot 100}{i}}$ for all $3 \leq i \leq 100$ to get,

$$\Leftrightarrow b_2^2 + (2k - 102)(x^{2 \cdot 3 \cdot \dots \cdot 100}) = b_{101}^{101}$$

We may now set $b_2 = 10 \cdot x^{\frac{2 \cdot 3 \cdot \dots \cdot 100}{2}}$,

$$\Leftrightarrow (2k - 2)(x^{2 \cdot 3 \cdot \dots \cdot 100}) = b_{101}^{101}$$

Note that since $k \geq 2$, we have $2k - 2 > 1$. We may now set,

$$x = (2k - 2)^m$$

Therefore,

$$(2k - 2)(x^{2 \cdot 3 \cdot \dots \cdot 100}) = (2k - 2)^{m \cdot 2 \cdot 3 \cdot \dots \cdot 100 + 1}$$

Since 101 is prime, we may find a m such that $101 \mid m \cdot 2 \cdot 3 \cdot \dots \cdot 100 + 1$ and we are done.

New: 46 clubs; 17.5 hours

Previous: 40 clubs; 17 hours

Total: 86 clubs; 34.5 hours